

SHUBHAM MADANE

Computer Science & Engineering | AI/ML | Generative AI | RAG | Computer Vision | Research

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M.Tech Computer Science & Engineering professional with a research and academic foundation across Artificial Intelligence, Machine Learning, Generative AI, Large Language Models, and Retrieval-Augmented Generation (RAG). Experienced in building RAG pipelines, multilingual NLP systems, and Computer Vision models using Python, alongside Java-based backend development. Published research on GAN-based data augmentation for agricultural disease detection, combining Generative AI with Computer Vision for real-world impact. Brings together AI/ML research, software development, and academic teaching contributions within an institutional CSE department setting.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, SQL

AI & Machine Learning: Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, Large Language Models, Natural Language Processing, Computer Vision, Neural Networks, CNN, GAN, DCGAN, Transfer Learning

Generative AI & RAG: Retrieval-Augmented Generation (RAG), LLMs, LangChain, FAISS, ChromaDB, Hugging Face, Sentence Transformers, Embeddings, Vector Databases, Semantic Search, Multilingual AI, Agentic AI, Prompt Engineering

Computer Vision: OpenCV, Image Processing, CNN, EfficientNetB0, Image Classification, Data Augmentation, GAN-based Synthetic Data Generation

Databases: Oracle

Frameworks & Tools: TensorFlow, Keras, NumPy, Pandas, Matplotlib, Streamlit, FastAPI, Flask, Git, GitHub, VS Code

PROFESSIONAL, TEACHING & ACADEMIC EXPERIENCE

Python Programming Laboratory | *Government College of Engineering, Chhatrapati Sambhajnagar*

- Conducted and supported hands-on Python programming laboratory sessions for Computer Science and Engineering students.
- Guided students in Python programming fundamentals, problem solving, debugging, and practical implementation.
- Assisted students in understanding programming concepts through practical demonstrations and laboratory exercises.
- Provided technical guidance and troubleshooting support during practical sessions.

Programming / Computer Science Teaching | *Private Coaching Classes*

- Taught Computer Science and programming concepts through structured classroom and practical instruction.
- Simplified complex programming concepts using examples, demonstrations, and hands-on exercises.
- Guided students in programming fundamentals, problem solving, and academic coursework.
- Supported students in preparation for academic examinations and practical assessments.

ACADEMIC & INSTITUTIONAL CONTRIBUTIONS

IEEE Student Chapter — Member | *Government College of Engineering, Chhatrapati Sambhajnagar*

- Contributed to IEEE Student Chapter activities and technical initiatives.
- Supported planning and execution of student-focused technical and academic activities.
- Collaborated with student and faculty teams in organizing technical events and initiatives.

Board of Studies (BoS) — Academic Contribution | *Government College of Engineering, Chhatrapati Sambhajnagar*

- Contributed to academic and departmental activities associated with the Board of Studies.
- Assisted in academic planning, curriculum-related activities, and departmental documentation.
- Supported academic coordination and institutional requirements.

Timetable & Academic Coordination | *Government College of Engineering, Chhatrapati Sambhajnagar*

- Contributed to departmental timetable planning and academic scheduling.
- Assisted in coordinating subjects, classes, laboratory sessions, and academic requirements.
- Supported smooth implementation of academic schedules and departmental coordination.

NAAC & Institutional Work | *Government College of Engineering, Chhatrapati Sambhajnagar*

- Contributed to NAAC-related academic and institutional documentation activities.
- Assisted in organizing and maintaining departmental and institutional records.
- Supported quality-assurance and accreditation-related institutional activities.
- Collaborated with faculty and departmental teams to complete assigned institutional requirements.

RESEARCH & PUBLICATIONS

Citrus Leaf Disease Detection Using GAN-Based Data Augmentation and Convolutional Neural Networks

International Journal of Advances in Signal and Image Sciences (IJASIS) | Published 9 March 2026

- Developed a citrus leaf disease classification framework integrating DCGAN-based synthetic image generation with EfficientNetB0 transfer learning.
 - Addressed limited agricultural training data by generating realistic synthetic images to improve model generalization.
 - Achieved **92.78% classification accuracy**, improving over the **86.42% baseline by 6.36 percentage points** (934 real images, 6 classes).
- <https://xlescience.org/index.php/IJASIS/article/view/1472>

A Comprehensive Review on CNN and GAN-Based Approaches for Citrus Leaf Disease Detection

Engineering Advances 2025 — Second International Conference (Conference Paper) | 29–30 December 2025

- Presented a comprehensive review of CNN- and GAN-based approaches for citrus leaf disease detection.
- Analyzed deep learning, image processing, and GAN-based data augmentation techniques for agricultural disease classification.
- Identified research challenges, limitations, and future directions for AI-driven plant disease detection.

SELECTED PROJECTS

Retrieval-Augmented Generation (RAG)-Based Multilingual Agricultural Advisory System for Indian Farmers

Python, LangChain, LLMs, RAG, FAISS, Hugging Face, Sentence Transformers, ChromaDB, Streamlit, FastAPI, Whisper, NLP

- Designed and developed a multilingual RAG-based agricultural advisory system using LLMs to generate context-aware, source-grounded responses.
- Implemented semantic document retrieval using FAISS and Sentence Transformers to retrieve relevant agricultural knowledge before response generation.
- Added multilingual interaction for **Marathi, Hindi, and English** to improve accessibility for diverse farming communities.
- Applied retrieval-grounded generation to reduce hallucinations and improve response reliability.
- Designed a scalable architecture using LangChain, vector databases, LLMs, and Streamlit/FastAPI components.

Citrus Leaf Disease Detection Using GAN-Based Data Augmentation and CNNs

Python, TensorFlow, Keras, DCGAN, CNN, EfficientNetB0, OpenCV, NumPy, Pandas, Matplotlib

- Developed a deep learning framework combining GAN-based synthetic image generation with CNN-based disease classification.
- Used DCGAN to generate realistic citrus leaf images and address limited agricultural training data.
- Applied EfficientNetB0 transfer learning for citrus disease classification, achieving 92.78% accuracy versus an 86.42% baseline.

Telemedicine Consultation Platform

Core Java, Oracle Database

- Designed a telemedicine platform concept enabling patients to connect with healthcare providers through virtual consultation.
- Planned features including appointment scheduling, secure messaging, medical record management, video consultation, and virtual prescriptions.
- Focused on improving healthcare accessibility for patients in remote areas.

EDUCATION

M.Tech — Computer Science & Engineering2026

Government College of Engineering, Chhatrapati Sambhajinagar • CGPA: 7.79/10

B.Tech — Computer Science & Engineering2023

Dr. Babasaheb Ambedkar Technological University • CGPA: 7.78/10

Higher Secondary Certificate (HSC)2018

Maharashtra State Board • 69.85%

Secondary School Certificate (SSC)2016

Maharashtra State Board • 76.00%

CERTIFICATIONS

- **Engineering Advances 2025 — Second International Conference** — Engineering Advances25, December 2025 (ID: 11272)
- **Build with AI — Workshop Participation** — GDG Chh. Sambhajinagar, July 2025
- **SQL (Intermediate)** — HackerRank, May 2024 (ID: 9e3df6afc548)
- **Java Programming Masterclass — Updated to Java 17** — Udemy, September 2023 (ID: UC-c2ab40a2-e117-4a9a-ab33-84073ab4d028)
- **Full Stack Java Developer** — Naresh i Technologies, January 2023
- **Oracle Advanced PL/SQL Developer Certified Professional** — MindLuster, October 2023 (ID: 12861401447)
- **Object Oriented Programming in Java** — Scaler, November 2023

RESEARCH INTERESTS

Generative Artificial Intelligence • Large Language Models • Retrieval-Augmented Generation • Multimodal AI • Vision-Language Models • Computer Vision • Natural Language Processing • Explainable AI • Agricultural AI • Precision Agriculture • Multilingual AI • Agentic AI

LANGUAGES

English • Marathi • Hindi